Journal of Physics D:
Applied Physics

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To cite this article: I L Semenov *et al* 2016 *J. Phys. D: Appl. Phys.* **49** 105204

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IOP Publishing Journal of Physics D: Applied Physics
J. Phys. D: Appl. Phys. **49** (2016) 105204 (12pp) doi:10.1088/0022-3727/49/10/105204

Numerical study of the anode boundary layer in atmospheric pressure arc discharges

I L Semenov[1], I V Krivtsun[2] and U Reisgen[3]

[1] Forschungsgruppe Komplexe Plasmen, Deutsches Zentrum für Luft- und Raumfahrt, Oberpfaffenhofen, Germany
[2] E.O. Paton Electric Welding Institute, Kiev 03680, Ukraine
[3] Welding and Joining Institute, RWTH Aachen University, Aachen 52062, Germany

E-mail: Igor.Semenov@dlr.de

Received 20 July 2015,
Accepted for publication 18 November 2015
Published 8 February 2016



Abstract

The anode boundary layer in atmospheric pressure arc discharges is studied numerically on the basis of the hydrodynamic (diffusion) equations for plasma components. The governing equations are formulated in a unified manner without the assumptions of thermal equilibrium, ionization equilibrium or quasi-neutrality. For comparison, a quasi-neutral model of the anode layer is also considered. The numerical computations are performed for an argon arc at typical values of the current density in anode layers (500–2000 A cm^{-2}). The results of numerical modelling show that the common collisionless model of the sheath fails to describe the sheath region for the problem under consideration. For this reason, a detailed analysis of the anode sheath is performed using the results of unified modelling. In addition, the distributions of plasma parameters in the anode layer are analysed and the basic characteristics of the layer (anode voltage drop, sheath voltage drop, anode layer thickness, sheath thickness, heat flux to the anode) are calculated. Our results are found to be in good agreement with the existing theoretical predictions and experimental data. The dependence of the anode layer characteristics on the current density is also discussed.

Keywords: anode boundary layer, atmospheric pressure arc discharges, numerical modelling, hydrodynamic equations

(Some figures may appear in colour only in the online journal)

1. Introduction

The near-electrode plasma layers of high pressure arc discharges have been the subject of intensive research for decades. Comprehensive reviews of both experimental and theoretical studies can be found in several recent publications on this topic [1–3]. As a rule, the cathode region is assumed to be more important than the anode one. For this reason, anode phenomena have attracted much less attention in the past.

In recent years, however, the anode region in atmospheric pressure arc discharges has received renewed interest owing to the continued development of arc welding technologies, such as gas–metal arc (GMA) welding, gas–tungsten arc (GTA) welding and hybrid (laser with GTA) welding processes. For example, the properties of the arc in GMA welding are still not well understood, mainly due to the effect of metal vapour on the anode boundary layer near a consumable wire electrode [4–10]. The influence of laser radiation on plasma properties in the anode region has also not been analysed in detail. In this view, further studies of the anode plasma layers seem to be of great importance for the field.

It is known that numerical modelling plays an important role for understanding physical processes in the near-electrode regions, since experimental studies of these thin non-equilibrium layers are difficult to perform and interpret. The anode boundary layer in atmospheric pressure discharges

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has been studied numerically using different models in many works [11–17]. We do not provide a detailed discussion of these works here and refer the reader to the reviews mentioned above [1, 2]. It should be emphasized, however, that the anode space-charge sheath has not been taken into account properly in previous studies. However, the analysis of the anode sheath seems to be an important issue, since the sheath might influence the properties of the whole anode layer. Therefore, it is highly desirable to model the anode region in a unified manner without any simplifying assumptions including quasi-neutrality. To the best of our knowledge, only one example of such a unified model has been reported in [18]. In this work the anode layer has been studied using the model proposed previously in [19] for modelling near-cathode plasma layers. However, the results presented in [18] are related to the case of very high pressure discharges (with pressures of about 100–200 bar) but not to the case of atmospheric pressure discharges.

In the present work we use the approach proposed in [19] to study the anode layer in atmospheric pressure arc discharges. One-dimensional unified modelling of the anode layer is carried out for an argon arc at different values of the current density in the range from 500–2000 A cm^{-2}. As was shown in [20] and mentioned in [2], this range is typical of the anode boundary layers of thermal arcs. The numerical modelling of the arc in the GMA welding process also shows that the current density near the anode wire lies in a similar range [4, 5]. For comparison, we also consider a quasi-neutral model of the anode layer supplemented with the common collisionless model of the sheath. Using the results of numerical modelling, we analyse the distribution of plasma parameters in the anode boundary layer and examine the anode sheath structure. In addition the basic characteristics of the anode layer, such as the anode voltage drop, sheath voltage drop, anode layer thickness, sheath thickness and total heat flux to the anode, are also presented and their dependence on the current density is discussed.

In view of the fact that we use a one-dimensional model to study the anode layer, a remark should be made about the applicability of this approach. As is known, the thickness of the anode layer may be quite substantial ($\sim$1 mm) compared, for example, to the typical length of the welding arcs. In this case, the radial variations of the plasma parameters in the arc column might noticeably affect the structure of the anode layer. For this reason, we should admit that the one-dimensional approach represents a strong simplification of the problem under consideration. On the other hand, a one-dimensional formulation enables us to extend the complexity of the physical model, e.g. by considering the anode layer without the assumption of quasi-neutrality. One further step in this direction might be to extend the approach proposed in [19] to the case of a multicomponent plasma with multiply charged ions. This step could be of importance for future studies needed to analyse the influence of metal vapour and laser radiation on the anode layer properties. However, such extensions of the physical model could be difficult to accomplish for more realistic geometry at the first stage. Furthermore, as is shown in the present study, the one-dimensional model of the anode layer can provide quite reasonable quantitative results.

The paper is structured as follows. In section 2 we present the governing equations, along with the corresponding boundary conditions, and describe briefly the numerical method employed to solve the formulated problem. The results of numerical computations are presented and discussed in section 3. The conclusions are summarized in section 4.

2. Model of the anode region

2.1. Basic equations

The approach employed in the present work is based mainly on the model proposed in [19]. However, we use a slightly different boundary condition for the ion number density at the anode surface. Moreover, we have also considered the quasi-neutral model of the anode region. Thus, for completeness, we present here the basic equations of the model and refer the reader to [19] for details.

Let us consider an anode boundary layer in an argon arc discharge under atmospheric pressure. The plasma in the anode layer is assumed to consist of atoms (a), singly charged ions (i) and electrons (e). The presence of multiply charged ions is neglected, since the plasma temperature in the bulk of the discharge is estimated to be not higher than 18 kK (for the considered range of current densities). The convective effects in the anode layer are also neglected. In addition, it is assumed that the heavy particles (ions and atoms) have a common temperature, T_h, which differs from the electron temperature, T_e (a two-temperature model).

As in [19], we consider a simple one-dimensional model of the anode region. The axis x of the Cartesian coordinate system is directed from the anode surface into the plasma and the anode is located in the plane $x = 0$. The governing equations of the model include the continuity, momentum and energy equations for plasma components. The continuity equations read as

$$\nabla J_\alpha = \omega_\alpha, \quad \alpha = \mathrm{e, i, a}, \tag{1}$$

where J_α is the number density flux, ω_α is the production rate of plasma particles due to ionization–recombination reactions and $\nabla = \mathrm{d}/\mathrm{d}x$. The number density flux is defined as $J_\alpha = n_\alpha \upsilon_\alpha$, where n_α and υ_α are the number density and velocity of the particles, respectively. In high pressure discharges the ionization is mainly driven by collisions between electrons and atoms and the dominant recombination mechanism is three-body recombination with an electron as the third body. In this case, the production rates are written as

$$\omega_e = \omega_i = -\omega_a = k_i n_a n_e - k_r n_e^2 n_i, \tag{2}$$

where k_i and k_r are the ionization and recombination rate constants, respectively. Adding the continuity equations for ions and atoms and subtracting the continuity equations for ions and electrons, we obtain the following relations:

$$J_i = -J_a, \quad J_i - J_e = j/e, \tag{3}$$

where j is the current density in the anode region (e is the elementary charge). Here the first relation reflects the conservation of heavy particles and the second relation reflects the continuity of the electric current in the anode region. Thus, relations (3) can be used instead of any two continuity equations.

The momentum equations for plasma particles can be written as (see [21])

$$-\nabla p_\alpha + n_\alpha z_\alpha E + \sum_\beta \nu_{\alpha\beta}\,\mu_{\alpha\beta}\,n_\alpha n_\beta\,(\upsilon_\beta - \upsilon_\alpha) - R_\alpha^{\mathrm{T}} = 0. \quad (4)$$

Here α, β denote the particle type; $p_\alpha = n_\alpha k T_\alpha$ is the partial pressure (k is the Boltzmann constant); $\mu_{\alpha\beta} = m_\alpha m_\beta/(m_\alpha + m_\beta)$ is the reduced mass ($m_{\alpha,\beta}$ denote the particles masses); z_α is the particle charge; E is the electric field; $\nu_{\alpha\beta}$ is the collision rate; and R_α^T is the thermal diffusion force given by

$$R_\alpha^{\mathrm{T}} = C_\alpha^{(\mathrm{e})} n_\alpha k \nabla T_\mathrm{e}, \quad (5)$$

where $C_\alpha^{(\mathrm{e})}$ is the thermal diffusion coefficient. Note that the thermal diffusion effect due to the heavy particles' temperature gradient is neglected in equation (5). The collision rate in equation (4) is related to the binary diffusion coefficient by

$$\nu_{\alpha\beta} = \frac{k T_{\alpha\beta} C_{\alpha\beta}}{n D_{\alpha\beta}\mu_{\alpha\beta}},$$

where $T_{\alpha\beta} = (m_\alpha T_\beta + m_\beta T_\alpha)/(m_\alpha + m_\beta)$ denotes the reduced temperature of the particles; $n = \sum_\alpha n_\alpha$ is the total number density; $D_{\alpha\beta}$ is the binary diffusion coefficient evaluated in the first approximation of the Chapman–Enskog method and $C_{\alpha\beta}$ are the coefficients accounting for approximations of higher orders.

Adding the momentum equation (4), we also obtain the following relation:

$$-\nabla p + e(n_\mathrm{i} - n_\mathrm{e})E = 0, \quad (6)$$

where $p = \sum_\alpha p_\alpha$ is the total pressure. The momentum equations are supplemented with the Poisson equation for the electric field:

$$\varepsilon_0 \nabla E = e(n_\mathrm{i} - n_\mathrm{e}) \quad (7)$$

(ε_0 is the electric constant). By combining equation (6) with equation (7) we obtain

$$\nabla(\,p - \varepsilon_0 E^2/2\,) = 0. \quad (8)$$

Equation (8) means that the total pressure in the system (including the electric field pressure) is constant across the anode region.

The energy equations for heavy particles and electrons can be written as

$$\nabla q_\mathrm{h} = \kappa_{\mathrm{eh}} n_\mathrm{e} k (T_\mathrm{e} - T_\mathrm{h}), \quad (9)$$

$$\nabla\left(q_\mathrm{e} + \frac{5}{2} J_\mathrm{e}\, k T_\mathrm{e}\right) = j\,E + \kappa_{\mathrm{eh}} n_\mathrm{e} k (T_\mathrm{h} - T_\mathrm{e}) - w_\mathrm{e}, \quad (10)$$

where q_e and q_h are the electron and heavy particles heat fluxes, respectively; κ_{eh} is the energy exchange frequency; and w_e is the electron energy losses due to inelastic processes. The energy exchange frequency is defined as

$$\kappa_{\mathrm{eh}} = \frac{3kT_\mathrm{e}}{m_\mathrm{i} n}\left(\frac{n_\mathrm{a}}{D_{\mathrm{ea}}} + \frac{n_\mathrm{i}}{D_{\mathrm{ei}}}\right),$$

and the electron energy losses are given by

$$w_\mathrm{e} = A_\mathrm{i}\,\omega_\mathrm{e} + w_{\mathrm{rad}},$$

where A_i is the energy of ionization and w_{rad} denotes the energy losses due to radiation. The heavy particles' heat flux is assumed to be caused only by the heat conduction, i.e. the effect inverse to the thermal diffusion is neglected. Thus, q_h is simply given by

$$q_\mathrm{h} = -\lambda_\mathrm{h} \nabla T_\mathrm{h}, \quad (11)$$

where λ_h is the heavy particles' thermal conductivity. In turn, both mechanisms are taken into account for the electrons, and the electron heat flux is given by

$$q_\mathrm{e} = -\lambda_\mathrm{e} \nabla T_\mathrm{e} + k\,T_\mathrm{e} n_\mathrm{e} \sum_{\alpha=\mathrm{i,a}} A_\alpha^{(\mathrm{e})}(\upsilon_\mathrm{e} - \upsilon_\alpha), \quad (12)$$

where λ_e is the electron heat conductivity and $A_\alpha^{(\mathrm{e})}$ are the kinetic coefficients.

Further, following the idea proposed in [19], we transform the governing equations to a set of four nonlinear differential equations of the second-order for the electron number density n_e, electric field E and temperatures T_h and T_e. To transform the governing equations we eliminate υ_a and υ_i from the momentum equation (4) for ions and electrons using relations (3). The resulting equations are written as

$$\begin{aligned} -\nabla(n_\alpha k T_\alpha) \pm e n_\alpha E - \gamma_\alpha J_\mathrm{e} \\ -\delta_\alpha\,(j/e) - C_\alpha^{(\mathrm{e})} n_\alpha k \nabla T_\mathrm{e} = 0, \quad \alpha = \mathrm{i,e}, \end{aligned} \quad (13)$$

where

$$\gamma_\mathrm{e} = \nu_{\mathrm{ea}}\,\mu_{\mathrm{ea}}\,(n_\mathrm{e} + n_\mathrm{a}) + \nu_{\mathrm{ei}}\,\mu_{\mathrm{ei}}\,(n_\mathrm{i} - n_\mathrm{e}),$$

$$\gamma_\mathrm{i} = \nu_{\mathrm{ia}}\,\mu_{\mathrm{ia}}\,(n_\mathrm{i} + n_\mathrm{a}) + \nu_{\mathrm{ei}}\,\mu_{\mathrm{ei}}\,(n_\mathrm{e} - n_\mathrm{i}),$$

$$\delta_\mathrm{e} = \nu_{\mathrm{ea}}\,\mu_{\mathrm{ea}}\,n_\mathrm{e} - \nu_{\mathrm{ei}}\,\mu_{\mathrm{ei}}\,n_\mathrm{e},$$

$$\delta_\mathrm{i} = \nu_{\mathrm{ia}}\,\mu_{\mathrm{ia}}\,(n_\mathrm{i} + n_\mathrm{a}) + \nu_{\mathrm{ei}}\,\mu_{\mathrm{ei}}\,n_\mathrm{e}.$$

By expressing J_e from equation (13) for electrons and substituting the resulting expression into the electron continuity equation we obtain the following equation for the electron number density:

$$\begin{aligned} \nabla\,[-\gamma_\mathrm{e}^{-1} k T_\mathrm{e} \nabla n_\mathrm{e} - \gamma_\mathrm{e}^{-1} n_\mathrm{e} e E - \gamma_\mathrm{e}^{-1}\delta_\mathrm{e}(j/e) \\ -\gamma_\mathrm{e}^{-1}\tilde{C}_\mathrm{e}^{\mathrm{e})} n_\mathrm{e}\, k \nabla T_\mathrm{e}] = k_\mathrm{i} n_\mathrm{a} n_\mathrm{e} - k_\mathrm{r} n_\mathrm{e}^2 n_\mathrm{i}, \end{aligned} \quad (14)$$

where $\tilde{C}_\mathrm{e}^{(\mathrm{e})} = 1 + C_\mathrm{e}^{(\mathrm{e})}$. The number densities of ions and atoms in equations (13) and (14) are also expressed as functions of n_e and E. The number density of ions is simply expressed using the Poisson equation. The number density of atoms is defined using condition (8), i.e. from the relation $p - \varepsilon_0 E^2/2 = \mathrm{const}$, where the constant is defined using the

values of the total pressure and electric field in the region of undisturbed plasma far from the anode (see also section 2.3).

Further, eliminating J_e from equation (13) and expressing n_i using the Poisson equation we obtain the equation for E. This equation reads

$$-\Theta_1(\varepsilon_0/e)\Delta E + \Theta_2\, n_e E + \Theta_3(\varepsilon_0/e)\nabla E + \Theta_4 n_e + \Theta_5 \nabla n_e + \Theta_6(j/e) = 0, \quad (15)$$

where $\Delta = \mathrm{d}^2/\mathrm{d}x^2$ and the coefficients are given by

$$\Theta_1 = \gamma_e k T_h, \quad \Theta_2 = e(\gamma_e + \gamma_i),$$

$$\Theta_3 = \gamma_e e E - \gamma_e\, k \nabla T_h - \gamma_e\, C_i^{(e)} k \nabla T_e,$$

$$\Theta_4 = [\,\gamma_i + \gamma_i C_e^{(e)} - \gamma_e C_i^{(e)}\,]\, k \nabla T_e - \gamma_e k \nabla T_h,$$

$$\Theta_5 = \gamma_i k T_e - \gamma_e k T_h, \quad \Theta_6 = \delta_e \gamma_i - \delta_i \gamma_e.$$

The equations for the temperatures T_h and T_e are obtained by substituting the expressions (11) and (12) into the energy equations (9) and (10), respectively. The resulting equation for T_h reads

$$\nabla(\lambda_h \nabla T_h) + \kappa_{eh} n_e k (T_e - T_h) = 0, \quad (16)$$

and the equation for T_e reads

$$\nabla(\lambda_e \nabla T_e) + j\, E + \kappa_{eh} n_e k (T_h - T_e) - A_i\, \omega_e - w_{rad} - \nabla\left[\frac{5}{2} J_e k T_e + k T_e n_e \sum_{\alpha=i,a} A_\alpha^{(e)} (\upsilon_e - \upsilon_\alpha)\right] = 0. \quad (17)$$

For comparison, we have also considered a quasi-neutral model of the anode region. This model, which is based on the constraint $n_e = n_i$, can be formulated as follows. First, the momentum equations (4) for ions and electrons are added and the ion number density flux J_i is expressed from the resulting equation with the help of relations (3). This expression is then substituted into the ion continuity equation to obtain the following equation for the ion number density:

$$\nabla\,[-\tilde{\gamma}_i^{-1} k (T_e + T_h)\nabla n_i + \tilde{\gamma}_i^{-1}\, \tilde{\delta}_i\, (j/e) - \tilde{\gamma}_i^{-1} n_i k \nabla T_h - \tilde{\gamma}_i^{-1} \tilde{C}_i^{(e)} n_i k \nabla T_e] = k_i n_a n_i - k_r n_i^3, \quad (18)$$

where $\tilde{C}_i^{(e)} = 1 + C_i^{(e)} + C_e^{(e)}$ and

$$\tilde{\gamma}_i = (\nu_{ia}\mu_{ia} + \nu_{ea}\mu_{ea})(n_i + n_a), \quad \tilde{\delta}_i = \nu_{ea}\mu_{ea} n_a.$$

The atom number density within the quasi-neutral model is defined using condition (6), which in this case means that $p = \mathrm{const}$. The electric field is calculated using the momentum equation (4) for electrons and the equations for temperatures T_h and T_e do not change.

2.2. Transport, kinetic and radiation coefficients

Transport, kinetic and radiation coefficients appearing in the set of governing equations have been defined using the same expressions as those used in [19]. In particular, the expressions for the coefficients $D_{\alpha\beta}$, $C_{\alpha\beta}$, $C_\alpha^{(e)}$, $A_\alpha^{(e)}$, λ_h and λ_e have been taken from appendix A of [19] (see equations (45)–(56) in [19]). The energy-averaged cross sections, which are required to evaluate the transport coefficients, were calculated as described below.

The energy-averaged cross section for momentum transfer in collisions between ions and atoms, $\bar{Q}_{ia}^{(1,1)}$, was calculated using data from [22]. Namely, we used the energy-dependent cross section for the charge-transfer process to evaluate $\bar{Q}_{ia}^{(1,1)}$ numerically as a function of T_h. The obtained results were then fitted using the formula $\bar{Q}_{ia}^{(1,1)} = 1.28 - 0.194\, \ln(T_h)^{0.9}$ (where T_h is in kK, $\bar{Q}_{ia}^{(1,1)}$ is in 10^{-18} m^2), which provides a quite accurate approximation for the cross-section $\bar{Q}_{ia}^{(1,1)}$ in the range from 500–25 000 K.

The energy-dependent cross section for momentum transfer in collisions between electrons and atoms was taken from the Biagi-v8.9 database (see [23]), which is available on the LXCat open-access website [24]. The corresponding energy-averaged cross section, $\bar{Q}_{ea}^{(1,1)}$, was then calculated numerically and tabulated as a function of T_e. The same procedure was used to evaluate the electron transport coefficients described by Lorentzian formulae (see, e.g. [25, 26]) in the case of partially ionized plasmas. Namely, we tabulated the correction coefficient C_{ea}, the thermal diffusion coefficient for electrons in the limit of partially ionized plasmas (coefficient C_{tdea} in equation (48) of [19]) and coefficient C_{term} in equation (51) of [19].

The energy-averaged cross section for momentum transfer in collisions between electrons and ions, $\bar{Q}_{ei}^{(1,1)}$, was calculated using the conventional formula based on the cut-off Coulomb potential (see, e.g. [26]):

$$\bar{Q}_{ei}^{(1,1)} = \frac{e^4 \ln\Lambda}{32\pi\varepsilon_0^2 (k T_e)^2},$$

where $\ln\Lambda$ is the Coulomb logarithm with $\Lambda = 1.24 \times 10^7\, T_e^{3/2}\, n_e^{-1/2}$ (here T_e is in K and n_e is in m^{-3}).

The energy-averaged cross sections $\bar{Q}_{aa}^{(2,2)}$, $\bar{Q}_{ii}^{(2,2)}$ and $\bar{Q}_{ia}^{(2,2)}$, which are required to evaluate λ_h, were defined as follows. The cross sections for atom–atom and ion–atom collisions were calculated using the fitting formulae proposed in [19] using the data of [22]: $\bar{Q}_{aa}^{(2,2)} = 1.12\, T_h^{-0.2}$, $\bar{Q}_{ia}^{(2,2)} = 3.6\, T_h^{-0.3}$ (where T_h is in K and the cross-sections are in 10^{-18} m^2). The cross-section $\bar{Q}_{ii}^{(2,2)}$ was obtained using the expression

$$\bar{Q}_{ii}^{(2,2)} = \frac{e^4 \ln\Lambda}{36\pi\varepsilon_0^2 (k T_h)^2},$$

which can be also found in [19, 27].

Further, the ionization and recombination rate constants were defined by means of the approach described in [28]. Namely, the ionization rate constant was represented as the sum of rate constants for direct and stepwise ionization, which were evaluated using the expressions presented in [28] (see equations (11)–(14) in [28]). The required data on parameters for argon atoms and ions were taken from [29] and the data on the ionization cross section were taken from [30]. The recombination rate was then calculated using the expression

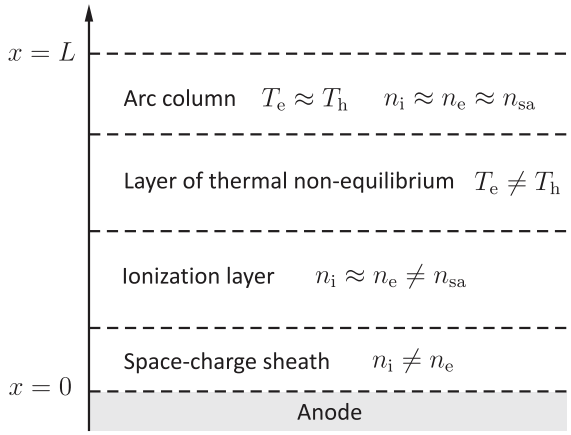


Figure 1. Schematic sketch of the anode boundary layer (see the description in the text).

$k_r = k_i(n_a/n_e^2)_{sa}$, where $(n_a/n_e^2)_{sa}$ denotes the ratio n_a/n_e^2 evaluated using the Saha equation.

Finally, the energy losses due to radiation were evaluated using the fitting formula proposed in [19] on the basis of the results of [31–33]. This formula reads

$$w_{rad} = 2.6 \times 10^{25} \frac{p}{T_e^{2.52}} \exp\left(-\frac{1.69 \times 10^5}{T_e}\right),$$

where p is in bars, T_e is in K and w_{rad} is in W m^{-3}.

2.3. Boundary conditions

Let us now formulate the boundary conditions used in the present study. For this purpose it might be useful to consider first the sketch of the anode layer shown in figure 1. As is known, the anode layer can be divided qualitatively into several sublayers corresponding to different kinds of perturbations introduced into the arc plasma by the anode. In particular, moving from the anode surface into the bulk plasma, one can distinguish the space-charge sheath, ionization layer and layer of thermal non-equilibrium (see figure 1). Here, the space-charge sheath is a thin layer, where the quasi-neutrality breaks down (i.e. $n_i \neq n_e$); the ionization layer is characterized by deviations from the ionization equilibrium (i.e. $n_i \approx n_e \neq n_{sa}$, where n_{sa} is the number density of the charged particles defined by the Saha equation); and the layer of thermal non-equilibrium is the region where T_e differs noticeably from T_h. Comprehensive description of these layers, along with estimates of their thicknesses, can be found, e.g. in the reviews [1–3]. Above the layer of thermal non-equilibrium there exists the region where the plasma is close to the local thermodynamic equilibrium. Within the one-dimensional model used in our study the plasma parameters in this region tend to uniform distributions at sufficiently large distances from the anode. In this case, the region of uniform plasma parameters represents a model of the arc column which can be defined by neglecting all spatial gradients in the governing equations described in section 2.1.

Taking into account the above discussion, we formulate the boundary-value problem for equations (14)–(17) on the interval $0 \leqslant x \leqslant L$, where the lower boundary, $x = 0$, coincides with the anode surface and the upper boundary, $x = L$, is placed far enough from the anode in the region of uniform plasma parameters. The boundary conditions imposed at $x = L$ are analogous to those used in [18, 19]. Namely, the number densities of plasma particles are found using the following equations:

$$k_i n_a n_e = k_r n_e^2 n_i, \tag{19}$$

$$p = n_a k T_h + n_i k T_h + n_e k T_e, \tag{20}$$

$$n_i = n_e, \tag{21}$$

where equation (19) means that the ionization is balanced (locally) by the recombination; equation (20) defines the total plasma pressure ($p = 1$ atm in our case); and equation (21) represents the quasi-neutrality condition. The number density fluxes and electric field can be found using equation (4) supplemented with relations (3). In particular, the electron flux is given by the expression

$$J_e = -\frac{j}{e} \frac{\delta_i + \delta_e}{\gamma_e + \gamma_i}, \tag{22}$$

and the electric field is given by

$$E = \frac{j}{e^2} \frac{\delta_i \gamma_e - \delta_e \gamma_i}{n_i \gamma_e + n_e \gamma_i}, \tag{23}$$

where the coefficients $\delta_{e,i}$ and $\gamma_{e,i}$ are defined below equation (13). Finally T_h and T_e far from the anode are defined using the conditions

$$T_h = T_e, \quad jE = w_{rad}(T_e). \tag{24}$$

Further, let us consider the boundary conditions imposed at the anode surface. We adopt the common assumption that electrons and ions recombine at the surface producing atoms and the remaining portion of the electrons is fully absorbed by the anode. The effect of the electron emission is neglected, since the surface temperature is assumed to be sufficiently low. In this case the boundary condition for the electron number density flux at the anode surface is given by

$$J_e = -\frac{n_e \bar{v}_e}{4}, \tag{25}$$

where $\bar{v}_e = \sqrt{8kT_e/\pi m_e}$ is the average speed of random (chaotic) thermal motion of the electrons. The right-hand side of this condition represents the flux of the electrons moving to the anode surface due to thermal motion. This flux can be obtained if we approximate the electron velocity distribution function at the anode surface by the (non-shifted) half-Maxwellian function which equals zero for the electrons moving from the surface (as the consequence of absorption). The left-hand side of the boundary condition (25) represents the electron flux evaluated within the hydrodynamic model. It is worth noting that the condition (25) is not strictly justified from the point of view of the kinetic theory of gases. Nevertheless, this

condition is frequently used in the modelling of both low and high pressure discharges.

On the other hand, the choice of the boundary condition for the electric field at the anode surface should be discussed in more detail. One possible way in this case is to assume that the ion number density at the anode surface has a fixed prescribed value, n_{is}. In this case, the boundary condition for E is simply defined using the Poisson equation (7):

$$\nabla E = (e/\varepsilon_0)(n_{is} - n_e). \tag{26}$$

For example in [19], it was assumed that $n_{is} = 0$ at the surface. Such a choice can lead to unrealistic values for the ion current density near the anode surface and, consequently, can influence the sheath structure. Thus, from the physical point of view, it is preferable to define the value of the ion number density flux at the anode surface. This question is discussed below.

It is known that due to absorption of ions there exists a so-called Knudsen non-equilibrium layer near the anode surface where the velocity distribution function of ions is discontinuous. The thickness of this layer is usually of the order of several mean free paths for ion–atom collisions. In this view, according to the preliminary estimates, this layer is expected to lie inside the sheath. The structure of the Knudsen layer is known to depend on the Mach number of the particles entering the layer [34] and, hence, on the particles' velocity if the temperature variation is neglected. In turn, the dynamics of ions in the sheath can be in many cases reasonably well described using the model of mobility-controlled ion motion [35]. Thus, when the presence of the Knudsen layer is neglected, the ion number density flux at the anode surface can be approximately given by

$$J_i = \frac{eEn_i}{\nu_{ia}\mu_{ia}n_a}. \tag{27}$$

The corresponding boundary condition for the electric field is then obtained as follows. The expressions (25) and (27) are substituted into the relation $J_i - J_e = j/e$ (current continuity condition) and the number density of the ions is expressed using the Poisson equation (7), i.e. we substitute $n_i = n_e + (\varepsilon_0/e)\nabla E$. The resulting condition for E reads as follows:

$$\frac{eEn_e}{\nu_{ia}\mu_{ia}n_a} + \frac{\varepsilon_0 E\,\nabla E}{\nu_{ia}\mu_{ia}n_a} + \frac{n_e\bar{\upsilon}_e}{4} - (j/e) = 0. \tag{28}$$

It should be noted that such an approach is expected to be justified for the case when $T_e \gg T_h$ and the ions are accelerated by the electric field in the sheath. Our computations showed that the last condition holds for the range of j we used. The electron temperature in the sheath was found to be only five times higher than the heavy particles' temperature. Nevertheless, we expect condition (28) to work reasonably well. Thus, in our computations we used expression (28) as the boundary condition for the electric field at the anode surface within the unified model. Further comparison between conditions (26) and (28) is also discussed in section 3.

For the temperatures T_h and T_e we use the same boundary conditions as those used in [19]. Namely, the heavy particles' temperature is assumed to be equal to the surface temperature T_s, i.e. the condition is

$$T_h = T_s. \tag{29}$$

And the boundary condition for the electron temperature at the anode surface reads as

$$-\frac{n_e\bar{\upsilon}_e}{4}\,2kT_e = \frac{5}{2}\,J_e\,kT_e + q_e, \tag{30}$$

where the effect of the electron thermal emission is neglected. The left-hand side of the condition (30) represents the energy flux transported by the electrons moving to the anode surface due to thermal motion. By analogy with the boundary condition (25), this flux can be obtained if we approximate the electron velocity distribution function at the anode surface by the (non-shifted) half-Maxwellian function. The right-hand side of the condition (30) represents the total energy flux evaluated within the hydrodynamic model.

Let us also discuss the boundary conditions for the quasi-neutral model of the anode region. In this case the boundary conditions for equations (16)–(18) at $x = L$ are the same as described above. The boundary conditions imposed at the point $x = 0$, which is assumed to coincide with the sheath edge, are described below. It should be noted that the computations based on the quasi-neutral model have been performed to emphasize the importance of the proper sheath description. In this view, we have used the common collisionless model for the sheath (see, e.g. [36]). Namely, the ion velocity at the sheath edge was assumed to be equal to the Bohm velocity $\upsilon_B = \sqrt{k(T_e + T_h)/m_i}$, i.e. the boundary condition for equation (18) was given by

$$J_i = -n_i\upsilon_B. \tag{31}$$

The boundary condition for T_h at $x = 0$ within the quasi-neutral model was defined using equation (29) and for T_e we used different prescribed values at the sheath edge in order to analyse the applicability of the common sheath model (see the discussion in section 3).

2.4. Method of numerical solution

Let us further describe briefly the numerical method which has been used to solve the boundary-value problem formulated in sections 2.1–2.3. For brevity, we describe here only the method used for solving the governing equations of the unified model, since the method used for solving the equations of the quasi-neutral model was virtually the same. The system of four nonlinear equations (14)–(17) was solved numerically by means of the following iterative procedure. First, the equations were linearized using Newton's method. In particular, the term n_eE in equations (14) and (15) was linearized with respect to n_e and E, the terms n_an_e and $n_e^2n_i$ in the production rate ω_e were linearized with respect to n_e and the term w_{rad} in equation (17) was linearized with respect to T_e. The other nonlinear terms were simply evaluated using the

plasma parameters from the previous iteration step. It should be also noted that the equations for n_e, E and T_h, T_e were decoupled. Namely, at each iteration step we first solved the coupled system of equations for n_e and E and then the coupled system of equations for the temperatures T_h and T_e.

The differential operators in equations (14)–(17) were discretized on a non-uniform mesh. This mesh was generated by applying a coordinate transform $x(\xi)$, where a new variable ξ was discretized on a uniform mesh on the interval [0, 1]. The transform function was defined as

$$x(\xi) = L\,(\sigma - 1)\frac{\bar{\sigma} - \bar{\sigma}^{1-\xi}}{1 + \bar{\sigma}^{1-\xi}},$$

where σ is the transform parameter; $\bar{\sigma} = (\sigma + 1)/(\sigma - 1)$; and L is the size of the computational domain (see p 544 in [37]). In our computations we used $\sigma = 1.01$.

To discretize the differential operators the governing equations were rewritten in terms of the new variable ξ and then the standard second-order finite difference scheme was applied to approximate the derivatives $\mathrm{d}/\mathrm{d}\xi$ and $\mathrm{d}^2/\mathrm{d}\xi^2$. It is worth noting that the geometric terms resulting from the applied coordinate transform were discretized using the same formulae. In this case the discrete analogue of the Laplace operator still equals zero for the linear function and no additional numerical noise is generated. The boundary conditions described in section 2.3 were also linearized and discretized using the second-order finite difference scheme.

The resulting system of linear equations obtained after discretization was solved using the conventional three diagonal matrix algorithm which was generalized to the case of coupled equations. In addition, a special procedure was applied to treat the boundary conditions near the anode surface. The iterations were stopped when the relative errors for the unknowns at each mesh point were below some threshold ($\sim 10^{-6}$).

3. Results and discussion

In this section we present the results of the numerical modelling which has been carried out on the basis of the anode region model described in section 2. As was mentioned in the introduction, the current density in our computations was varied in the range from 500–2000 A cm^{-2}. For this range of current densities, the plasma temperature in the bulk of the discharge varies between 13 and 18 kK. The anode surface temperature T_s was not varied in our study and was chosen to be equal to 1 kK. The computations have been performed for both the unified and quasi-neutral models of the anode boundary layer.

First, in order to emphasize the importance of proper description of the anode sheath, we have performed computations on the basis of the quasi-neutral model with different prescribed values of T_e at the sheath edge. As it was described in section 2.3, in this case we used an assumption based on the common sheath model (see [36]), i.e. the ion velocity at the sheath edge was assumed to be equal to the Bohm velocity. To analyse the applicability of the common sheath model let us consider the ratio of J_e to the chaotic electron flux $J_e^{(ch)} = -n_e\bar{v}_e/4$ at the sheath edge. In table 1 we present the values of this ratio calculated within the quasi-neutral model at different current densities j and different T_e at the sheath edge. Note that the results in table 1 are given for relatively low values of T_e. The reason is that for higher values of T_e the gradient of the electron temperature at the surface becomes negative and, hence, the electron heat conduction flux becomes positive (the heat flows out of the sheath). It seems to be an artificial effect caused by the imposed boundary condition. Moreover, the results of a unified modelling showed that the electron temperature near the anode surface lies in the range from 5–6 kK.

Table 1. The ratio of the electron number density flux (J_e) to the chaotic electron flux ($J_e^{(ch)}$) at the sheath edge computed within the quasi-neutral model of the anode layer. The results are presented for different current densities (j) and different electron temperatures (T_e) at the sheath edge.

j, A cm^{-2}	$J_e/J_e^{(ch)}$		
	$T_e = 4$ kK	$T_e = 5$ kK	$T_e = 6$ kK
700	5.45	4.67	4.12
1000	3.69	3.12	2.73
1500	2.16	1.81	1.57
2000	1.41	1.18	1.03

One can see from table 1 that $|J_e|$ is higher than $|J_e^{(ch)}|$ for the considered parameters. On the other hand, as was mentioned, for example, in [38], the common collisionless model of the sheath is justified when $|J_e|$ is much smaller than $|J_e^{(ch)}|$. Namely, if this condition is not satisfied, the assumption of Boltzmann distributed electrons inside the sheath and condition $v_i = -v_B$ at the sheath edge become questionable. It should also be emphasized that the standard sheath model is assumed to describe the electron-repelling sheath. In this case, to ensure that the sign of the sheath voltage drop obtained within the standard model is consistent with this assumption, the condition $|J_e| < |J_e^{(ch)}|$ at the sheath edge has to be satisfied as well. Thus, our results show that the quasi-neutral model of the anode layer supplemented with the boundary condition (31) (i.e. $v_i = -v_B$) at the sheath edge gives contradictory results. This problem indicates that a unified modelling of the anode boundary layer without the assumption of quasi-neutrality is indeed of importance.

A possible explanation for why the common sheath model does not work for the problem under consideration might be fact that under some conditions an electron-attracting anode sheath can form. For example, as was found in [39, 40], such conditions can be realized in Hall thrusters. Note, however, that the analogy in this case is not complete because Hall thrusters operate at low pressures. The other important point to be noted is that the anode sheath is not fully collisionless. Our estimations showed that the ion–atom mean free path near the anode surface is of the order of 0.1 μm, while the sheath thickness is of the order of 1 μm (see table 3 below). This means that the ion–atom collisions might affect the structure of the anode sheath. On the other hand, we have tried to account for

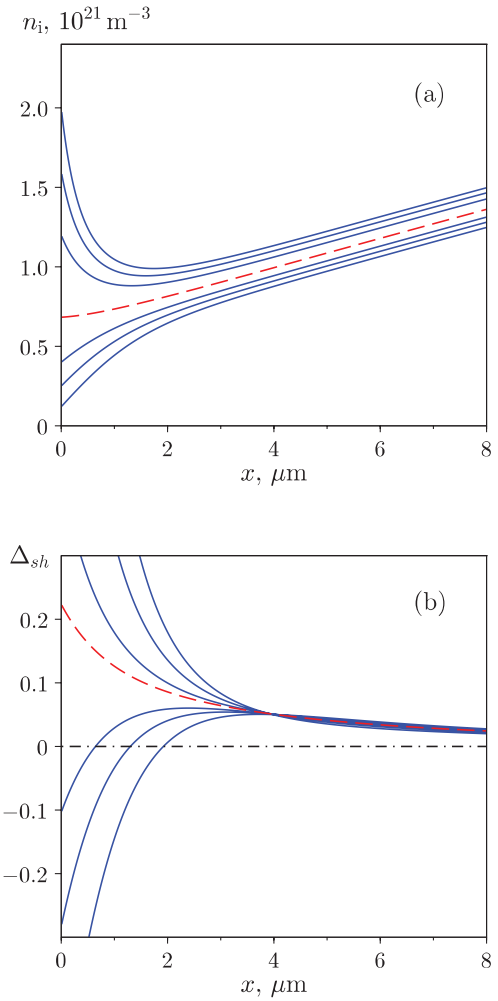


Figure 2. Distributions of the ion number density (a) and relative charge separation (b) near the anode surface for the case $j = 1000$ A cm^{-2}. The results are obtained within the unified model of the anode layer using two different boundary conditions for the electric field at the anode surface. The solid lines show the solutions obtained using the boundary condition (26) with different prescribed values of n_{is} and the dashed line shows the solution obtained using the boundary condition (28).

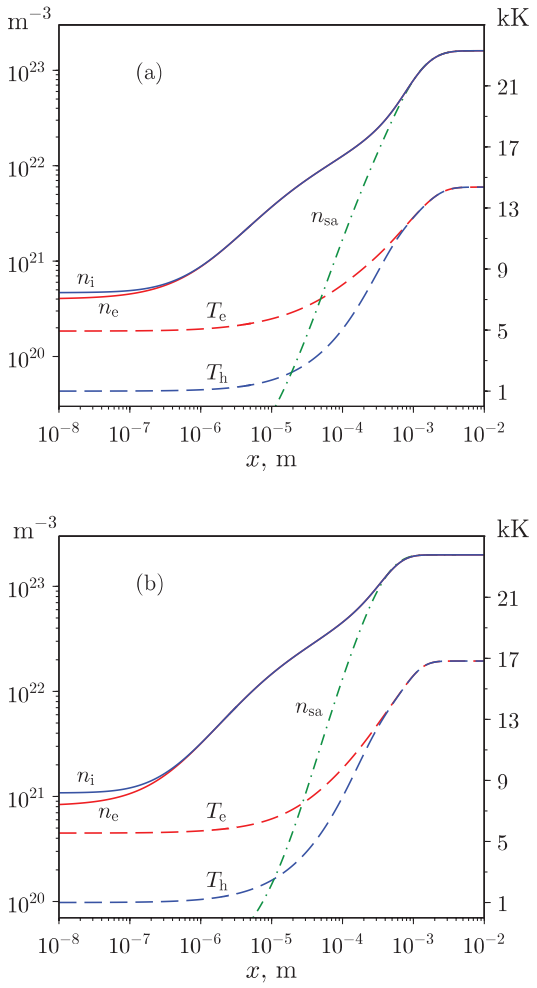


Figure 3. Distributions of the plasma parameters in the anode boundary layer for two different values of the current density: $j = 700$ A cm^{-2} (a) and $j = 1500$ A cm^{-2} (b). The solid lines show the number densities of the electrons (n_e) and ions (n_i). The dashed lines show the temperatures of the electrons (T_e) and heavy particles (T_h). The dashed-dotted lines show the charged particle number density (n_{sa}) calculated by means of the Saha equation and quasi-neutrality condition.

the effect of collisions by using the modified Bohm velocity at the sheath edge [41–43], but it was found that this approach also leads to contradictory results.

Let us now consider the results obtained on the basis of the unified model of the anode layer. However, before we present the main part of our results, let us briefly discuss the boundary condition imposed for the electric field at the anode surface in the unified model (see section 2.3 for the details). In figure 2 we show the distributions of the ion number density and relative charge separation in the sheath obtained for the case $j = 1000$ A cm^{-2} using two different boundary conditions: the boundary condition (26) with different prescribed values of n_{is} and the boundary condition (28). The relative charge separation is given by $\Delta_{sh} = (n_i - n_e)/n_{es}$, where n_{es} is the electron number density at $x = 0$. It can be seen from figure 2 that the boundary condition with the prescribed n_{is} can noticeably affect the solution near the surface. For example, the charge separation changes sign in the sheath at small values of n_{is} (see figure 2(b)). It is likely to be an artificial effect, since the charge separation near the sheath edge is positive. On the other hand, the boundary condition (28) seems to work reasonably well, because the solution obtained with this condition provides a uniform transition between the region near the sheath edge and the anode surface.

Further, let us discuss the main part of the results obtained using the unified model. In figure 3 we show the distributions of plasma parameters in the anode layer for the cases $j = 700$ A cm^{-2} and $j = 1500$ A cm^{-2}. The parameters shown are the number densities of the ions and electrons and the temperatures of the electrons and heavy particles. In addition, we also show the number density of the charged particles, n_{sa}, calculated by means of the Saha equation and quasi-neutrality condition. In figure 3 one can see the region of uniform plasma parameters representing the solution of equations (19)–(24). In this region which starts at $x \sim 1$ mm the plasma is close to the local thermodynamic equilibrium, i.e. $n_i \approx n_e \approx n_{sa}$ and $T_e \approx T_h$. The anode boundary layer can be observed at $x \lesssim 1$

Table 2. The electron temperature at the anode surface and in the undisturbed plasma for different values of the current density.

j, A cm^{-2}	T_e, kK (anode surface)	T_e, kK (undisturbed plasma)
500	4.69	13.53
700	4.93	14.37
1000	5.20	15.40
1500	5.59	16.82
2000	5.99	18.04

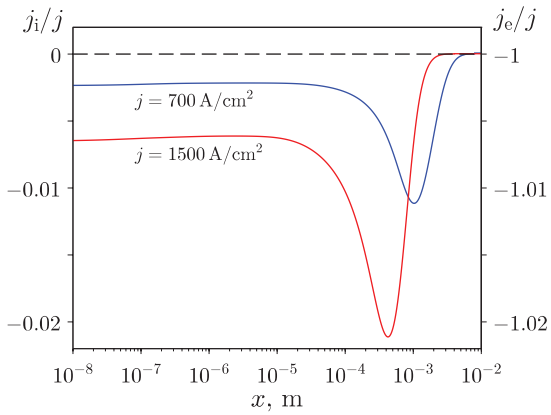


Figure 4. Distributions of the ion (j_i) and electron (j_e) current densities normalized to j for the cases $j = 700$ A cm^{-2} and $j = 1500$ A cm^{-2}. Note that due to relation $J_i - J_e = j/e$ the distributions of j_i and j_e differ only by the constant j.

mm. In particular, one can distinguish the layer of thermal non-equilibrium ($T_e > T_h$), the ionization layer ($n_{i,e} > n_{sa}$) and the space-charge sheath ($n_i > n_e$). The detailed description of these non-equilibrium sublayers can be found in [1–3, 19].

As will be shown later, the general trend is that the thickness of the anode layer decreases with increasing current density. This is simply explained by the fact that the plasma temperature in the bulk of the discharge increases with j. In turn, this leads to an increase in the ioniziation and energy exchange rates and, consequently, to a decrease in the thickness of the ionization and thermal non-equilibrium layers. It is also worth noting that the plasma temperature in the bulk of the discharge at $j = 1500$ A cm^{-2} is close to that obtained in [4, 5] at similar values of the current density near the wire anode in GMA welding. Another interesting point to note is that the electron temperature at the anode surface changes slightly as the current density increases. For clarity, the values of the electron temperature at the anode surface and in the undisturbed plasma are given in table 2 for different j.

In addition to the results shown in figure 3 we also present in figure 4 the distributions of the electron and ion current densities for $j = 700$ A cm^{-2} and $j = 1500$ A cm^{-2}. The electron and ion current densities are defined as $j_e = eJ_e$ and $j_i = eJ_i$, respectively. Note also that due to relation $J_i - J_e = j/e$, the distributions of j_i and j_e differ only by the constant j. The distribution of j_i shown in figure 4 agrees qualitatively with that presented in figure 6 of [11]. Namely, one can observe a non-monotonic behaviour of j_i, which can be explained as follows. According to the solution of equations (19)–(24), the ion

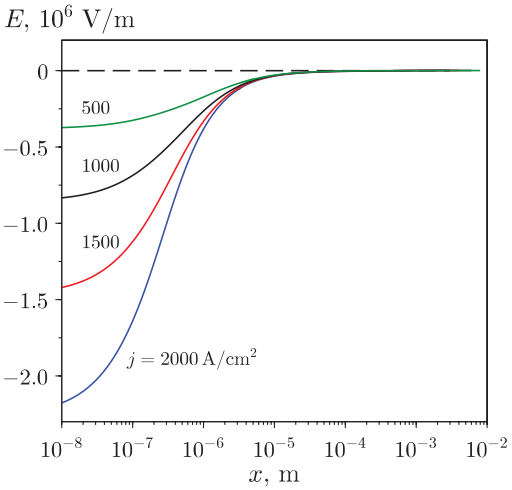


Figure 5. Distributions of the electric field in the anode layer for different values of the current density.

current density in the region of undisturbed plasma is positive, i.e. directed outward from the anode. Inside the anode layer the ion current density changes sign and increases towards the anode due to absorption of ions and electrons at the anode surface. On the other hand, near the anode surface the ion current density decreases owing to the increase of atoms number density which leads to the reduction of the ion mobility (see also the discussion in [11]).

Further, we show in figure 5 the distributions of the electric field in the anode layer for different values of the current density. As it follows from the solution of equations (19)–(24), the electric field is positive in the region where the plasma parameters are uniform. On the other hand, inside the anode layer the electric field associated with the density gradients (presheath field) plays a dominant role and the total electric field changes sign. It is worth noting that, in contrast to the case of very high pressure discharges studied in [18], the behaviour of the electric field shown in figure 5 is monotonic near the point of sign change. The influence of the anode sheath structure on the electric field strength near the anode surface is also visible in figure 5. Namely, as the current density decreases, the charge separation inside the sheath becomes smaller which leads to a decrease in the electric field gradient and strength near the anode surface. This tendency might be explained by taking into account that the electron temperature near the anode surface also decreases with decreasing j (see table 2).

To supplement the discussion of the anode sheath, we show in figure 6 the distributions of the ion velocity near the the anode surface for different values of j. The dashed line in this figure marks a rough estimate of the anode sheath thickness and the dashed-dotted line shows the representative value of the Bohm velocity calculated at $T_h = T_s$, $T_e = 5$ kK. It can be seen from figure 6 that the ion velocity at the sheath edge is much lower than the Bohm velocity. This result can partially explain why the the quasi-neutral model of the anode layer gives contradictory results when it is combined with the standard sheath model assuming that $v_i = -v_B$ at the sheath edge.

In the final part of this section we present the basic characteristics of the anode boundary layer computed using the

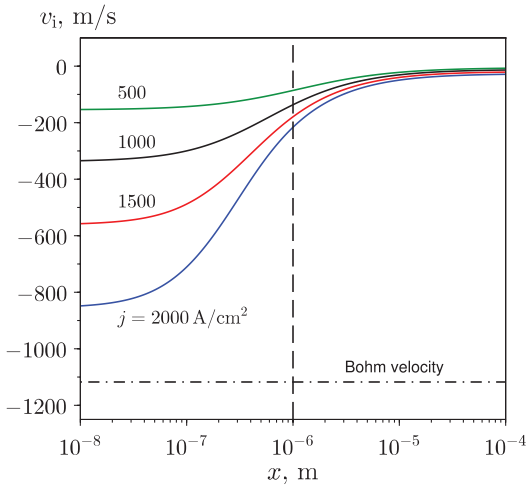


Figure 6. Distributions of the ion velocity in the anode layer for different values of the current density. The dashed line marks a rough estimate of the sheath thickness. The dashed-dotted line shows the Bohm velocity calculated at $T_h = T_s$, $T_e = 5$ kK.

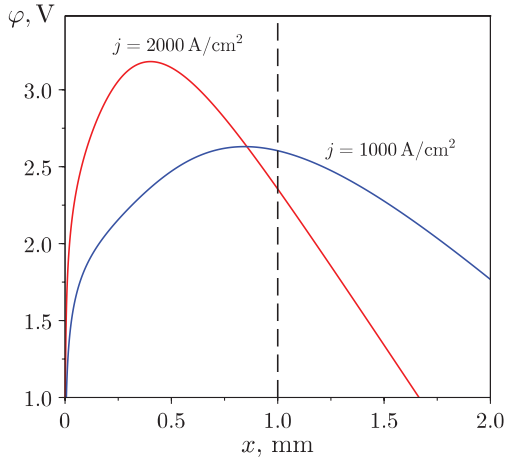


Figure 7. Distributions of the electric potential in the anode layer for $j = 1000$ A cm^{-2} and $j = 2000$ A cm^{-2}. The dashed line marks the point $x = 1$ mm.

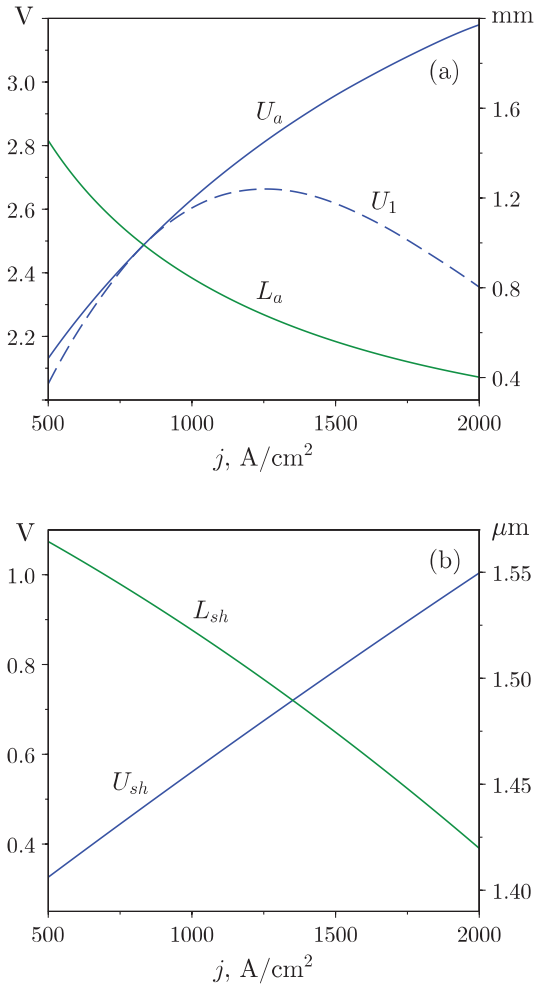


Figure 8. (a) The dependence of the anode voltage drops and anode layer thickness on the current density. (b) The dependence of the sheath voltage drop and sheath thickness on the current density. The definitions of U_a, U_1, U_{sh} and L_a, L_{sh} are given in the text.

results of the unified modelling. First, let us consider the voltage drop in the anode layer. In order to clarify the definition of the anode voltage drop we show in figure 7 the distributions of the electric potential in the anode layer for $j = 1000$ A cm^{-2} and $j = 2000$ A cm^{-2}. The zero of the potential is chosen to be at the anode surface. Note also that the sheath is not resolved in figure 7. One can see from figure 7 that the electric potential reaches a maximum value at some point in the layer. At this point the electric field associated with the density gradients (the pre-sheath field) is equal to the electric field associated with the electric current (see also [14]). It is natural to assume that the boundary of the anode layer is defined by the zero point of the electric field. Thus, in what follows we define the anode voltage drop, U_a, as the maximum value of the potential in the anode region and the thickness of the anode layer, L_a, as the distance between the surface and the point of the potential maximum. It should be noted, however, that the experimental measurements of the voltage drops in the anode layer are usually performed at some fixed point near the surface [2, 44]. In this view, we consider also the voltage drop U_1 measured between the surface and the point $x = 1$ mm. In addition, we consider the voltage drop across the anode sheath, U_{sh}, and the anode sheath thickness, L_{sh}. Here U_{sh} is defined as the voltage drop measured between the surface and the point $x = L_{sh}$ where the relative charge separation Δ_{sh} equals 1%.

The dependencies of L_a, U_a and U_1 on the current density are shown in figure 8(a). It can be seen that the obtained values of U_a and U_1 are close to those measured experimentally [44]. One can also see that U_a uniformly increases with j, while U_1 has a non-monotonic dependence on j with a decreasing part. The behaviour of U_1 can be understood taking into account the electric potential distributions shown in figure 7. When L_a is close to 1 mm, the voltage drops U_a and U_1 are close to each other and U_1 increases with j. However, at higher values of j the point $x = 1$ mm lies outside the anode layer and U_1 decreases, as can be seen from figure 7. As was mentioned in [2], the anode voltage drops measured experimentally can both increase and decrease with j depending on the arc attachment mode. The results shown in figure 8(a) might have a connection to these observations. In any case these results

Table 3. The values of different voltage drops in the anode layer, the anode layer thickness and sheath thickness at different current densities. The definitions of U_a, U_1, U_{sh} and L_a, L_{sh} are given in the text.

j, A cm^{-2}	U_a, V	U_1, V	U_{sh},V	L_a, mm	L_{sh}, μm
500	2.13	2.05	0.33	1.46	1.564
700	2.36	2.34	0.42	1.14	1.555
1000	2.63	2.60	0.56	0.84	1.525
1500	2.96	2.62	0.79	0.56	1.474
2000	3.18	2.35	1.00	0.40	1.420

indicate that the location of the point where the voltage drop is measured can also play a role in interpreting the experimental results. Another point that can be seen in figure 8(a) is that the thickness of the anode layer decreases with increasing j. As was mentioned previously, this is explained by the fact that the plasma temperature in the bulk of the discharge increases. The same reason seems to explain the increase in U_a with j. The dependencies of U_{sh} and L_{sh} on the current density are shown in figure 8(b). One can see that U_{sh} and L_{sh} follow the same tendencies as U_a and L_a. However, in contrast to L_a, the sheath thickness decreases slightly as j increases. It can be also seen that U_{sh} can reach sufficiently small values as the current density decreases. This observation shows that the anode sheath in atmospheric pressure discharges might have similar features as those discussed in [39, 40]. For completeness, we summarize in table 3 the values of U_a, U_1, U_{sh} and L_a, L_{sh} at different current densities.

Finally, let us consider the heat flux coming from the plasma to the anode surface. The total anode heat flux q_a can be expressed as

$$q_a = q_{a1} + q_{a2} + q_{a3} + q_{a4} + q_{a5}, \tag{32}$$

where

$$q_{a1} = \lambda_e \nabla T_e, \quad q_{a2} = \lambda_h \nabla T_h, \tag{33}$$

are the heat fluxes caused by heat conduction;

$$q_{a3} = -\frac{5}{2} J_e k T_e - k T_e n_e \sum_{\alpha=i,a} A_\alpha^{(e)} (v_e - v_\alpha), \tag{34}$$

is the electron heat flux caused by convection; and

$$q_{a4} = -J_i A_i, \quad q_{a5} = (J_i - J_e) A_f, \tag{35}$$

are the heat fluxes caused by the recombination of ions and absorption of electrons at the anode surface, respectively. In equation (35) A_f is the work function of the anode material. In our study we have used the value $A_f = 4.5$ eV which is close to the case of the steel anode. The dependencies of the total anode heat flux and heat fluxes defined by equations (33)–(35) on the current density are shown in figure 9. The results presented in figure 9 are in agreement, by an order of magnitude, with the previous results of numerical modelling (see figures 8 and 9 in [14]) and available experimental data (see figure 30 in [2]). The relative contributions of different terms in equation (32) also agree with the results of [14, 44]. Namely, the main contribution comes from the absorption of the electrons at the anode surface. It is evident from equation (35) that this

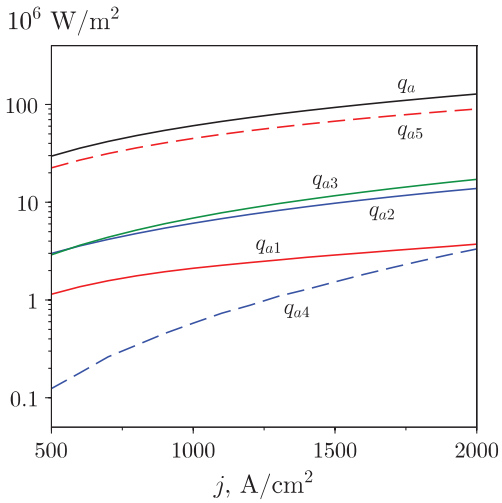


Figure 9. The dependencies of the total anode heat flux q_a and heat fluxes defined by equations (33)–(35) on the current density.

Table 4. The values of the total anode heat flux q_a and heat fluxes defined by equations (33)–(35) for different values of the current density. The heat fluxes are presented in the units 10^6 W m^{-2}.

j, A cm^{-2}	q_a	q_{a1}	q_{a2}	q_{a3}	q_{a4}	q_{a5}
500	21.5	0.8	2.2	2.1	0.1	16.3
700	30.3	1.1	3.0	3.2	0.2	22.8
1000	43.9	1.5	4.4	5.0	0.4	32.6
1500	67.6	2.1	7.1	8.5	1.1	48.9
2000	92.7	2.7	10.0	12.4	2.4	65.2

term increases with j. Other important terms are the heat conduction flux of the heavy particles and the electron heat flux caused by convection. Both these fluxes increase with j due to increasing temperature gradients between the undisturbed plasma and the anode surface. The electron conduction heat flux and heat flux caused by the recombination of ions seem to be of minor importance. Nevertheless, it can be seen that the heat flux caused by the recombination of ions increases noticeably with j. Taking into account equation (35), it can be easily explained by an increase in the ion velocity at the anode surface (see figure 6). For completeness, the values of the total anode heat flux and heat fluxes defined by equations (33)–(35) are summarized in table 4 at different current densities.

4. Conclusions

To conclude, we have studied the anode boundary layer in atmospheric pressure argon discharges by applying the approach proposed previously in [19]. The model of the anode layer is based on the hydrodynamic equations for plasma components. The distinctive feature of the model is that the governing equations are formulated in a unified manner without the simplifying assumptions of thermal equilibrium, ionization equilibrium or quasi-neutrality. This approach allows one to analyse, in particular, the structure of the space-charge sheath near the anode surface. The governing equations have been solved numerically and the computations have

been performed for typical values of the current density. For comparison, computations based on the quasi-neutral model of the anode layer supplemented with the common collisionless model of the sheath have been performed as well.

The results of the numerical modelling show that the common model of the sheath fails to describe the anode sheath region for the problem under consideration. Namely, if it is assumed that the ion velocity at the sheath edge equals the Bohm velocity, the results obtained within the quasi-neutral model are not consistent with the basic assumptions of the sheath model. For this reason, the detailed analysis of the anode sheath has been performed using the results of a unified modelling. It is shown that the ion velocity at the sheath edge is usually much lower than the typical Bohm velocity. The influence of the current density on the electric field distribution in the sheath is also demonstrated.

In addition, the results of unified modelling have been used to analyse the distributions of plasma parameters in the anode layer. It is shown, for example, that the electron temperature near the anode surface changes slightly in the range from 5–6 kK with increasing current density, while the plasma temperature in the bulk of the discharge increases noticeably in the range from 13–18 kK. The basic characteristics of the anode layer have also been calculated. Namely, the values of the anode voltage drop, anode sheath drop, anode layer thickness, sheath thickness and the total heat flux to the anode are presented. Moreover, the dependence of these characteristics on the current density is discussed and the relative contributions of different processes to the formation of the anode heat flux are analysed.

Finally, we want to mention that, despite the simplicity of the model we employed, the present work can provide a first step toward the modelling of the anode layer under conditions typical of technological processes such as GMA arc welding and hybrid laser–arc welding. For this purpose, the effects of anode evaporation and laser radiation on the anode layer properties have to be taken into account. These tasks are left for future work.

Acknowledgments

Financial support by the German Research Foundation via the project ‘Self-consistent model for Laser-TIG and Laser-Plasma hybrid welding in heat transfer mode’ (DFG RE 2755/20-1) is gratefully acknowledged.

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